

Legendary Status in Pokémon: A Data-Driven Classification and Clustering Analysis

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Abstract

Legendary Pokémon represent an extreme minority (46 out of 721, 6.38%), making them a compelling case study in **imbalanced classification**. This work aims to investigate whether legendary status can be **predicted** strictly from combat and morphological traits (RQ1) and whether Pokémon naturally **cluster** around this label or around alternative properties (RQ2). For RQ1, eight algorithms were evaluated under a hybrid SMOTE and cost-sensitive strategy, validated via Cross-Validation and Holdout testing. While standard metrics like Accuracy and ROC-AUC proved misleading, robust evaluation metrics identified the **Multi-Layer Perceptron**, **XGBoost** and **Logistic Regression** as the top balanced performers, whereas tree ensembles collapsed toward majority-class predictions. For RQ2, *k*-Means clustering, applied without the legendary label, identified **three** natural groups rather than two, driven primarily by overall combat power and by Body Mass Index. Legendary Pokémon share their statistical space with top-tier standard Pokémon, demonstrating that, absent an explicit label, a true Legendary is statistically **indistinguishable** from a comparably strong standard counterpart.

Keywords

Machine Learning — Imbalanced Classification — Cost-Sensitive Learning — k-Means Clustering — Pokémon

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Introduction

The Pokémon franchise has shaped pop culture since its introduction in the 1990s. Not only children, but also teenagers and adults have been captivated by these creatures. The primary objective of this research is to explore the underlying patterns of Legendary Pokémon through both supervised and unsupervised learning techniques. Specifically, this work aims to answer the following research questions:

1. **Supervised Analysis (RQ1):** Is it possible to accurately predict a Pokémon's legendary status based strictly on its combat capabilities and intrinsic physical traits, deliberately excluding descriptive gameplay mechanics to prevent target leakage?
2. **Unsupervised Analysis (RQ2):** Do Pokémon naturally cluster based on their legendary classification, or do they group around other intrinsic characteristics?

1. Dataset description

The dataset chosen to conduct the analysis is *Pokémon for Data Mining and Machine Learning*, available on the Kaggle Platform [1] and last updated in 2017. It consists of 721 statistical units and the following 23 features¹:

¹While some variables like *Number* or *Generation* are read as numerical integers by the software, they are classified here according to their true statistical scale of measurement to ensure methodological rigor.

- **Number:** (categorical - nominal). Pokémon ID in the Pokédex.
- **Name:** (categorical - nominal). Name of the Pokémon.
- **Type 1:** (categorical - nominal). Primary type.
- **Type 2:** (categorical - nominal). Secondary type, if present.
- **Total:** (numerical - ratio). Sum of all base stats (HP, Attack, Defense, Special Attack, Special Defense, and Speed).
- **HP:** (numerical - ratio). Base Health Points.
- **Attack:** (numerical - ratio). Base Attack.
- **Defense:** (numerical - ratio). Base Defense.
- **Sp_Atk:** (numerical - ratio). Base Special Attack.
- **Sp_Def:** (numerical - ratio). Base Special Defense.
- **Speed:** (numerical - ratio). Base Speed.
- **Generation:** (categorical - ordinal). Generation in which the Pokémon was introduced.
- **isLegendary:** (categorical - binary). Indicates whether the Pokémon is Legendary.
- **Color:** (categorical - nominal). Color of the Pokémon according to the Pokédex.
- **hasGender:** (categorical - binary). Indicates whether the Pokémon can be classified as female or male.
- **Pr_male:** (numerical - ratio). Given that the Pokémon has a gender, the probability of it being male (the probability of being female is complementary).
- **Egg_Group 1:** (categorical - nominal). Primary Egg Group of the Pokémon.
- **Egg_Group 2:** (categorical - nominal). Secondary Egg Group of the Pokémon, if present.
- **hasMegaEvolution:** (categorical - binary). Indicates whether the Pokémon is able to Mega-evolve.
- **Height_m:** (numerical - ratio). Height of the Pokémon, in meters.
- **Weight_kg:** (numerical - ratio). Weight of the Pokémon, in kilograms.
- **Catch_Rate:** (numerical - ratio). Internal base value determining capture difficulty (higher values indicate easier capture).
- **Body_Style:** (categorical - nominal). Body Style according to the Pokédex.

1.1 Exploratory Data Analysis (EDA)

Before proceeding with feature engineering, a preliminary inspection of the dataset was conducted to assess the structure and distribution of the available variables.

The target variable *isLegendary* is severely unbalanced, with only 6.38% of the Pokédex falling into this category (46 out of 721 Pokémon), as shown in Figure 1. This imbalance represents a central methodological challenge addressed throughout the supervised learning phase, motivating the stratified sampling and cost-sensitive strategies implemented in Section 3.1.

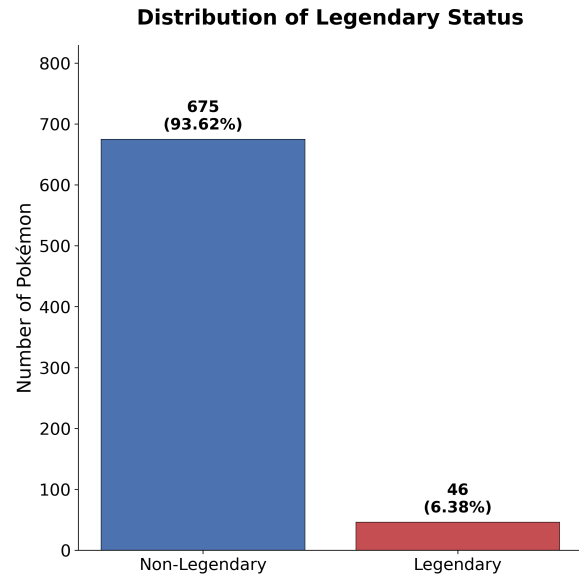


Figure 1. Distribution of *isLegendary*

2. Feature Engineering

2.1 Missing Values Handling & Binarization

An initial inspection of the dataset revealed missing values in *Type_2*, *Egg_Group_2*, and *Pr_Male*. For categorical features, missingness simply indicated the absence of a secondary attribute, therefore null records were imputed with the string label "NULL". For *Pr_Male*, missing values corresponded to genderless creatures and were encoded as -1 to preserve this information without distorting the probability scale. However, cleaning these features was performed solely for the sake of data integrity and completeness, as they were ultimately excluded from the feature set due to their purely descriptive nature.

Furthermore, *Rule Engine* nodes were employed to binarize boolean variables *isLegendary*, *hasGender* and *hasMegaEvolution* into discrete 0/1 indicators, ensuring compatibility with the mathematical requirements of the selected machine learning algorithms.

2.2 Feature Creation & Variable Selection

Consistently with the exclusion of descriptive gameplay mechanics, all purely categorical or identification attributes (*Num-*

ber, Name, Type_1, Type_2, Color, Egg_Group_1, Egg_Group_2, Body_Style and Generation) were excluded a priori, as none of them represented combat capability or morphology. The remaining candidate variables were then subjected to data-driven screening to detect residual redundancy or hidden data leakage not evident from conceptual considerations alone.

The physical build of each Pokémon was aggregated into a single continuous feature by computing the **Body Mass Index** ($BMI = \frac{Weight_{kg}}{(Height_m)^2}$), subsequently dropping the original weight and height statistics. This choice also prevented data leakage, as these raw dimensions were highly correlated with *isLegendary*.

Afterwards, the distributions of *Catch_Rate*, *hasGender*, *Pr_Male*, and *hasMegaEvolution* were inspected to determine whether legendary creatures were systematically assigned specific values. The analysis revealed that legendary Pokémon were always assigned one of two fixed *Catch_Rate* values. In fact, 91.30% had a value of 3, while the remaining 8.70% had a value of 45 [2]. Furthermore, 100% of legendaries were genderless [3] and 41 out of 46 could not Mega-evolve. Consequently, these variables were excluded from the final feature set, as they represented hardcoded game design choices tied directly to the target variable.

The remaining continuous variables were used to compute a Pearson correlation matrix to evaluate linear dependencies. The *Total* variable exhibited correlation coefficients above 0.5 with all base stats, as it represents their exact sum. It was therefore dropped in favor of its individual components.

Ultimately, the final subset of quantitative predictors retained to model the target variable resulted in: *HP*, *Attack*, *Defense*, *Sp_Atk*, *Sp_Def*, *Speed*, and *BMI*.

3. RQ1: Supervised Learning for Legendary Prediction

3.1 Data Preprocessing and Imbalance Strategy

Because the chosen machine learning algorithms rely on different mathematical assumptions, the preprocessing pipeline implemented specific transformations to supply models with their optimal data format. Furthermore, to prevent algorithms from collapsing into trivial majority-class predictions under severe class imbalance, a hybrid preprocessing and resampling strategy was implemented:

- **Normalization:** Distance-based and gradient-based algorithms are highly sensitive to feature scales. Therefore, continuous variables were standardized using Z-score normalization. For the Multi-Layer Perceptron, Min-Max scaling ([0, 1]) was explicitly applied to facilitate activation function convergence. Tree-based algorithms, which are scale-invariant, were processed through the Z-score normalizer merely to maintain pipeline homogeneity.

- **Synthetic Oversampling (SMOTE):** For models operating in a geometric feature space (e.g., Random Forest, XGBoost, Multi-Layer Perceptron, K-Nearest Neighbors, Logistic Regression), the Synthetic Minority Oversampling Technique (SMOTE) [4] was applied strictly to the training folds. This increased the density of the minority class in the feature space without inducing data leakage into validation folds.
- **The Naive Bayes Exception and Cost-Sensitive Learning:** Because SMOTE generates synthetic instances through linear interpolation, it creates artificial correlations that violate the conditional independence assumption of Naive Bayes. Consequently, Naive Bayes bypassed the SMOTE node and was trained on the original distribution. To compensate for the lack of oversampling, a *Cost-Sensitive* approach was integrated specifically for this algorithm. A higher misclassification penalty was assigned to False Negatives (penalty = 10) compared to False Positives (penalty = 1). This custom cost matrix modified the effective decision threshold, penalizing models that achieved high overall accuracy merely by ignoring the rare class.

3.2 Evaluated Models

To capture both linear and non-linear patterns, a diverse set of machine learning algorithms was evaluated:

- **Logistic Regression**
- **Decision Tree**
- **Random Forest**
- **Tree Ensemble**
- **eXtreme Gradient Boosting (XGBoost)**
- **K-Nearest Neighbors (KNN)**
- **Naive Bayes**
- **Multi-Layer Perceptron (MLP)** trained via **Resilient Backpropagation (Rprop)**

3.3 Validation Protocol and Hyperparameter Tuning

To rigorously evaluate model performance and optimize architectural parameters, a multi-stage validation pipeline was designed.

During hyperparameter optimization, a stratified 70/30 Train-Validation split was utilized and kept entirely separate from the 5-fold Cross-Validation later used for final model evaluation. SMOTE oversampling was applied exclusively to the training partition. An exhaustive Brute Force search strategy was then executed across the hyperparameter space: for each evaluated configuration, the algorithm was trained on the 70% split and tested on the held-out 30% validation set. Optimal parameters were systematically identified by maximizing the minority-class *F1-Score*, preventing overfitting during tuning.

Algorithms relying on direct probabilistic calculations or closed-form solutions, such as Naive Bayes and standard Logistic Regression, did not require structural tuning and were evaluated in their standard formulations. The tuned hyperparameters and their final selected values are summarized in Table 1.

Table 1. Hyperparameter tuning configurations via Brute Force search.

Algorithm	Tuned Parameters (Selected Value)
Decision Tree	Minimum records per node (10)
Random Forest	Number of Trees (70), Max Depth (20)
Tree Ensemble	Number of Trees (50), Max Depth (20)
XGBoost	Scale position weight (11), Max Depth (3), Learning rate (0.2), Number of rounds (50)
K-Nearest Neighbor (KNN)	Number of Neighbors k (17)
Multi-Layer Perceptron (MLP Rprop)	Hidden Layers (4), Neurons per layer (48), Maximum iterations (300)

Once optimal hyperparameters were determined, final predictive performance was evaluated. The tuned configurations were deployed within a stratified **5-fold Cross-Validation** architecture to assess metric stability across data partitions, followed by a final **Holdout** evaluation to benchmark generalization on unseen data. A fold count of $k = 5$ was deliberately chosen over higher alternatives (e.g., $k = 10$) to ensure each test fold retained a sufficient absolute number of positive instances (Legendary Pokémon) to yield stable metric estimates.

3.4 Metric Definitions and Selection Strategy

Because the Holdout test set preserves the original imbalanced distribution of the *isLegendary* class, standard evaluation metrics can yield misleading conclusions. A naive baseline predicting the majority class ("*Not Legendary*") for all instances would achieve artificially high performance. To prevent this bias, models were assessed using both standard baselines and metrics robust to class imbalance [5]:

- **Accuracy:** It is the ratio of correctly predicted observations to total observations ($\frac{TP+TN}{Total}$). While reported as a baseline, it is fundamentally flawed with imbalanced datasets, as the dominant negative class inflates the score.
- **Recall (Sensitivity):** It corresponds to the *True Positive Rate* ($\frac{TP}{TP+FN}$). Recall is critical for this research, as it measures the model's ability to identify rare Legendary Pokémon without missing them.
- **Specificity:** Also defined as the *True Negative Rate* ($\frac{TN}{TN+FP}$), measuring the proportion of standard Pokémon correctly identified, it reflects resistance to false alarms.
- **F1-Score:** It is measured as the harmonic mean of

Precision and Recall ($2 \times \frac{Precision \times Recall}{Precision + Recall}$). It provides a balanced assessment that prevents optimizing one error type at the expense of another. While Recall remains crucial, optimizing it in isolation rewards degenerate models that classify all instances as positive. For this reason, F1-Score served as the primary optimization objective.

- **Receiver Operating Characteristic (ROC) Curve and Area Under the Curve (AUC):** ROC Curves plot *True Positive Rate* against *False Positive Rate* across varying decision thresholds. Under severe imbalance, ROC-AUC remains overly optimistic because the vast number of True Negatives suppresses the False Positive Rate.
- **Precision-Recall (PR) Curve and Average Precision (AP):** PR Curves plot Precision directly against Recall, ignoring True Negatives entirely. The resulting area under the curve, quantified as *Average Precision (AP)*, provides a reliable benchmark of minority-class discriminative power.

3.5 Results and Discussion

The empirical evaluation of the models across both the 5-fold cross-validation and the final Holdout test set revealed significant variations in their ability to handle the severely imbalanced *isLegendary* classification task. Baseline assessments demonstrated that standard evaluation metrics were heavily skewed by the overwhelming presence of standard Pokémon (True Negatives). In fact, overall Accuracy consistently exceeded 0.87 and ROC-AUC scores remained optimistically high across all architectures.

Having empirically confirmed the misleading nature of these standard metrics, the comparative analysis focused strictly on measurements robust to severe class imbalance: F1-Score, Recall, Specificity and Average Precision (AP), derived from the Precision-Recall curve. The comprehensive performance metrics, divided by evaluation strategy (Holdout and 5-fold Cross-Validation), are detailed in Table 2 and Table 3.

Table 2. Performance metrics: 5-Fold Cross-Validation (Mean Values).

Model	Rec.	Prec.	Spec.	F1	Acc.	Cohen's κ	AP	ROC-AUC
Logistic Regression	0.807	0.752	0.981	0.765	0.969	0.749	0.845	0.988
Decision Tree	0.569	0.622	0.973	0.575	0.947	0.548	0.564	0.868
Random Forest	0.262	0.793	0.994	0.360	0.947	0.343	0.670	0.967
Tree Ensemble	0.264	0.793	0.994	0.363	0.947	0.345	0.636	0.952
XGBoost	0.804	0.718	0.976	0.751	0.965	0.732	0.782	0.982
K-Nearest Neighbors	0.676	0.730	0.982	0.695	0.963	0.675	0.678	0.963
Naive Bayes	0.958	0.364	0.881	0.524	0.886	0.475	0.779	0.968
Multi-Layer Perceptron (Rprop)	0.789	0.760	0.984	0.771	0.971	0.755	0.829	0.973

3.5.1 Cross-Validation vs. Holdout Generalization

A direct comparison between the averaged metrics of the 5-fold Cross-Validation (CV) and the Holdout evaluation confirmed the structural stability of the chosen hyperparameters. The models demonstrated consistent behavior across both testing frameworks, proving that the tuning phase successfully

Table 3. Performance metrics: Holdout Evaluation.

Model	Rec.	Prec.	Spec.	F1	Acc.	Cohen's κ	AP	ROC-AUC
Logistic Regression	0.786	0.611	0.966	0.688	0.954	0.663	0.815	0.985
Decision Tree	0.643	0.750	0.985	0.692	0.963	0.673	0.686	0.928
Random Forest	0.286	1.000	1.000	0.444	0.954	0.428	0.773	0.969
Tree Ensemble	0.286	1.000	1.000	0.444	0.954	0.428	0.728	0.950
XGBoost	0.929	0.812	0.985	0.867	0.982	0.857	0.888	0.981
K-Nearest Neighbors	0.714	0.769	0.985	0.741	0.968	0.724	0.765	0.952
Naive Bayes	1.000	0.438	0.911	0.609	0.917	0.570	0.850	0.990
Multi-Layer Perceptron (Rprop)	0.786	0.917	0.995	0.846	0.982	0.836	0.908	0.994

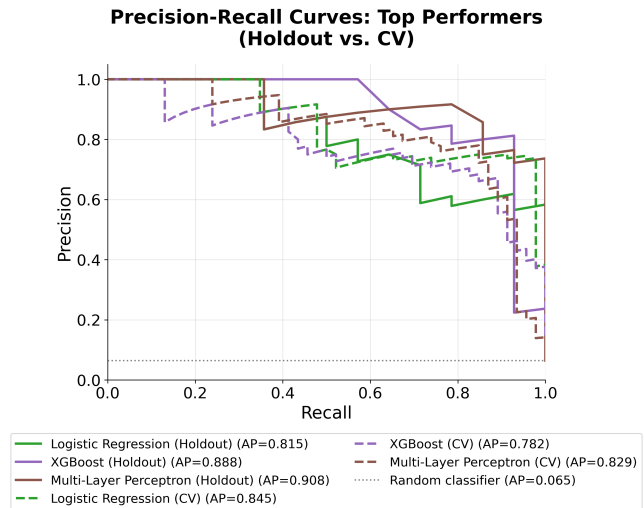
prevented overfitting on the validation partitions. Architectures that performed well on the CV folds generalized effectively to unseen Holdout data. Conversely, the architectural limitations observed during CV were mirrored in the Holdout test, confirming they represented systemic behaviors rather than artifacts of a specific partition.

3.5.2 Top Performers: Multi-Layer Perceptron, XGBoost and Logistic Regression

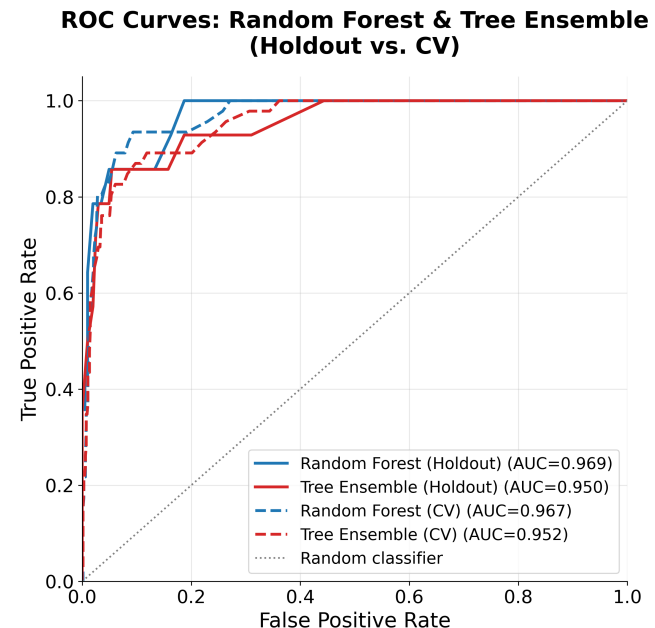
No single architecture dominated uniformly across both evaluation protocols and all metrics. Instead, three models consistently occupied the top tier, each exhibiting distinct strengths. The Multi-Layer Perceptron (MLP) trained via Rprop proved to be the most consistent performer overall, achieving the highest ROC-AUC in both CV (0.973) and Holdout (0.994), the highest Holdout AP (0.908) and the highest CV F1-Score (0.771), while maintaining a solid Recall (0.789 in CV, 0.786 in Holdout). This confirmed the neural network's superior capability to map non-linear decision boundaries for the minority class without collapsing into majority-class predictions. XGBoost exhibited the largest generalization gain in F1-Score, rising from 0.751 in CV to the highest overall Holdout F1-Score (0.867), surpassing the Multi-Layer Perceptron (0.846) while registering the highest Holdout Recall (0.929) among all balanced models. This result empirically validated the effectiveness of the *Scale Position Weight* parameter as a direct mathematical countermeasure to class imbalance. Logistic Regression achieved the highest CV AP (0.845), slightly edging out the neural network (0.829) and remained highly competitive in Holdout (0.815), demonstrating that a well-calibrated linear baseline can rival more complex architectures on this task. Together, these three models represented the only architectures capable of sustaining high Recall (above 0.78 in Holdout) without suffering a collapse in Precision, establishing them as viable candidates for practical deployment.

3.5.3 The Cost of Regularization

Random Forest and Tree Ensemble both exhibited severe performance degradation in identifying the minority class across both evaluation strategies. Random Forest registered a critically low Recall in both the averaged CV (0.262) and the Holdout evaluation (0.286), despite achieving perfect Precision (1.000) in Holdout. The resulting F1-Score (0.444) exposed this apparent perfection as an artifact of extreme conservatism rather than genuine discriminative capability. This collapse is a direct result of the aggressive regularization applied during hyperparameter tuning, as heavily restricting maximum tree depth over-constrained the models, damaging


Figure 2. Precision-Recall curves for the top three models (Multi-Layer Perceptron, XGBoost, Logistic Regression).

their structural capacity to isolate rare Legendary Pokémon and causing them to default to majority-class predictions to minimize overall loss. Nevertheless, both models retained misleadingly high ROC-AUC scores (Figure 3), with values exceeding 0.95, reinforcing the argument made in Section 3.4 that ROC-AUC fails to expose practical inadequacy on the minority class under severe imbalance.


Figure 3. ROC curves for Random Forest and Tree Ensemble across both the Holdout set (solid lines) and the 5-fold Cross-Validation (dashed lines).

3.5.4 The Cost-Sensitive Effect: Naive Bayes

The Naive Bayes classifier demonstrated a typical response to the Cost-Sensitive framework, illustrating why Recall alone

is an insufficient selection criterion. By inflating the misclassification penalty for *False Negatives*, the model adopted an ultra-conservative decision boundary, achieving near-perfect Recall across both evaluation schemes (0.958 in CV, 1.000 in Holdout). However, this absolute sensitivity compromised Precision severely (0.364 in CV, 0.438 in Holdout), yielding a low F1-Score (0.524 in CV, 0.609 in Holdout) driven by high false-positive rates. Naïve Bayes thus exhibited the inverse failure mode of the tree ensembles, as rather than ignoring the minority class it systematically over-predicted it.

4. RQ2: Unsupervised Analysis and Cluster Profiling

To address the second research question and determine whether Pokémon naturally group based on their *isLegendary* classification or other intrinsic properties, a *k*-Means clustering algorithm was applied to the selected feature subset (all components of *Base_Stats* and *BMI*), with the *isLegendary* label excluded during training to ensure unbiased grouping. The optimal number of clusters was determined by evaluating the Silhouette Coefficient for $k \in [2, 11]$. The search identified $k = 3$ as the optimal configuration (Silhouette Coefficient = 0.275), outperforming the intuitive binary split ($k = 2$, Silhouette Coefficient = 0.263) that would have emerged had legendary status been the primary discriminant in the data. This result provided the first indication that Pokémon group around alternative structural properties.

4.1 Dimensionality and Visual Separation

The resulting 3-cluster structure, obtained on the original 7-dimensional feature space, is visualized in two dimensions using Principal Component Analysis (PCA) in Figure 4. This dimensionality reduction technique served strictly as a visualization aid and played no role in the clustering process itself, which was conducted directly on the original features.

The first two principal components jointly explain 61.0% of the total variance (PC1: 32.3%, PC2: 28.7%). Examining the component loadings, PC1 loads positively and uniformly across all six base combat statistics (loadings between 0.33 and 0.46), with a negligible contribution from BMI (−0.057). It can therefore be interpreted as a proxy for overall combat power. This axis separates weaker standard Pokémon (*cluster_0*, blue) on the left from elite Pokémon (*cluster_2*, green) on the right. PC2, by contrast, loads most strongly on BMI (0.641), Defense (0.524) and inversely on Speed (−0.481), isolating a distinct structural archetype (*cluster_1*, orange) along the vertical axis.

Projecting legendary status onto this geometric space (Figure 4, right) reveals that Legendary Pokémon do not form an isolated cluster. Instead, they occupy the upper extreme of PC1, visually coinciding with the domain populated by *cluster_2*, while remaining largely absent from the high-PC2 space characterizing *cluster_1*. Section 4.3 quantifies this overlap in terms of cluster composition.

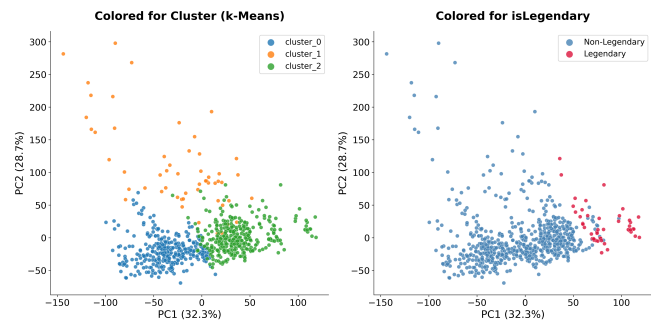


Figure 4. PCA scatter plots showing *k*-Means cluster assignment (left) and actual legendary status (right). Legendaries do not form an isolated group but merge with high-performing standard Pokémon.

4.2 Feature Importance and the Morphology Factor

To determine the driving forces behind this separation, an ANOVA F-statistic analysis was conducted across the cluster features. As shown in Figure 5, Body Mass Index (BMI) emerged as the most discriminative feature by a wide margin, followed by physical Defense and HP. The accompanying boxplot confirms the morphological profile of the divergent *cluster_1*, which contains Pokémon exhibiting substantially higher BMI values than the rest of the dataset.

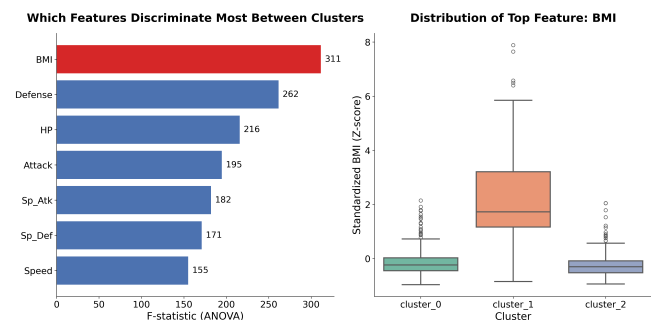


Figure 5. ANOVA F-statistic ranking of cluster features (left) and distribution of the top discriminator, BMI (right). *cluster_1* is clearly defined by extreme morphological weight rather than combat stats alone.

4.3 Legendary Composition and Mean Base Stats in Clusters

The internal composition of these clusters definitively resolves the second research question. Figure 6 shows that Legendary Pokémon are entirely absent from *cluster_0*. They are primarily concentrated in the high-stats *cluster_2* (12.4% composition) and the heavy-weight *cluster_1* (5.6% composition).

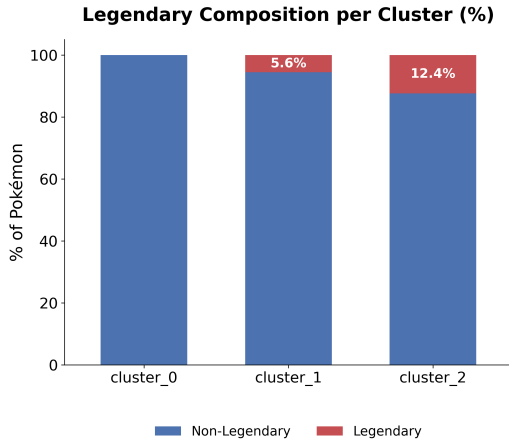


Figure 6. Legendary composition per cluster. `cluster_0` contains no Legendaries, while `cluster_2` and `cluster_1` contain 12.4% and 5.6% respectively.

Crucially, isolating non-legendary Pokémon (Figure 7) clarifies why the algorithm grouped them with Legendaries. Standard Pokémon assigned to `cluster_2` exhibit an average base stats approaching 500, substantially exceeding the overall non-legendary baseline of 404.

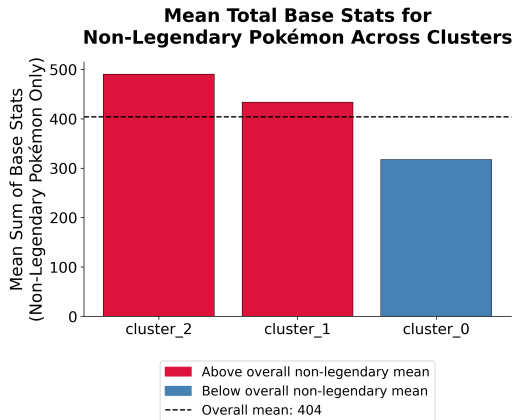


Figure 7. Mean sum of base stats for strictly non-legendary Pokémon across clusters, showing that the algorithm groups elite standard Pokémon with Legendaries due to shared numerical superiority.

5. Conclusion

This research investigated the classification and clustering of Legendary Pokémon based strictly on intrinsic combat capabilities and morphological traits.

For the supervised classification task (RQ1, Section 3), addressing the severe class imbalance was essential, especially given that conventional metrics such as Accuracy and ROC-AUC yielded misleading conclusions. No single architecture dominated uniformly across all robust metrics: Multi-Layer Perceptron (MLP), XGBoost and Logistic Regression consis-

tently formed the top performing tier, each presenting unique trade-offs, and together constituted the only models capable of sustaining high Recall without significant Precision loss. Conversely, Random Forest and Tree Ensemble collapsed under heavy regularization, defaulting to majority-class predictions. Naive Bayes, constrained by cost-sensitive weighting, maximized Recall at the direct expense of Precision, confirming that Recall alone cannot serve as a reliable optimization objective and validating the F1-Score as the appropriate tuning criterion.

Regarding the unsupervised analysis (RQ2, Section 4), the results confirm that Pokémon do not naturally cluster by legendary status. Evaluating the data without the *isLegendary* label revealed an optimal 3-cluster structure driven by total combat capabilities and morphological traits, specifically Body Mass Index. Because top-tier standard Pokémon match the basic stat totals of true Legendaries, both groups occupy overlapping statistical profiles. Therefore, if there is no explicit label, Legendary Pokémon remain indistinguishable from elite standard creatures.

6. Future Work

Future work could leverage Natural Language Processing (NLP) to analyze Pokédex descriptions, combining textual lore with the numerical data used here. Additionally, expanding the scope to include subsequent Pokémon generations would allow researchers to assess the temporal stability of these combat and morphological distributions, as the overall sample size grows.

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